

Trainings ▾

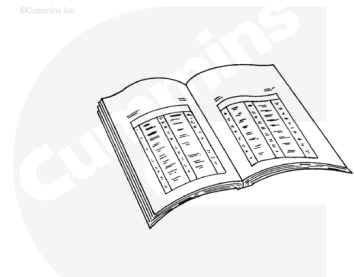
Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



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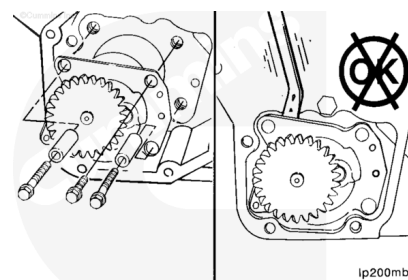
Note : The water pump must be removed to remove the water pump idler gear.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Remove the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)
- Remove the water pump idler gear. Refer to Procedure 001-040 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-040-tr.html>)

Remove

⚠ CAUTION ⚠

Do not pry on the gear or pump flange. This can result in damage to the pump.



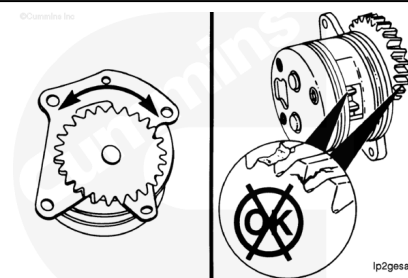
Remove the three mounting capscrews and spacers.

Use both hands to pull, or insert, two capscrews (M8 X 1.25) into the puller holes to push the pump from the engine.

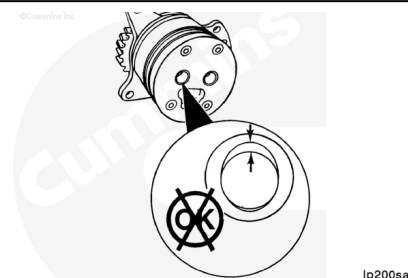
Inspect for Reuse

Turn the main drive gear by hand to check the gears for freedom of rotation.

Inspect the gears for cracked, worn, or broken teeth.

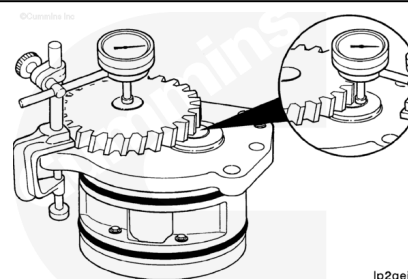


Inspect the rear cover plate shaft bores for excessive wear or discoloration due to overheating or shaft seizure.



Measure the driveshaft and driven shaft end clearance.

Driveshaft End Clearance		
mm		in
0.064	MIN	0.0025
0.269	MAX	0.0106

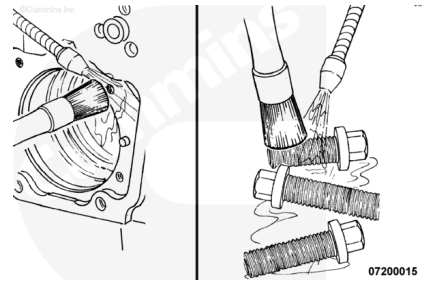


If the gears are damaged, the shaft does **not** turn freely by hand, or the end clearance is **not** within the specifications given, the oil pump **must** be replaced.

Install

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

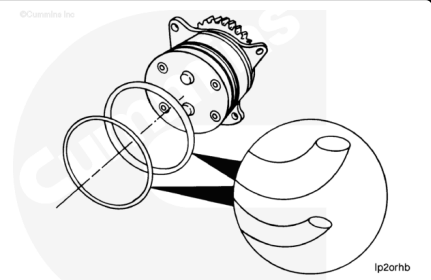
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the threaded lubricating pump mounting holes in the cylinder block and capscrew threads with solvent. Dry with compressed air.

Do **not** lubricate the two oil pump body o-rings. The o-rings will swell, and the pump can **not** be installed.

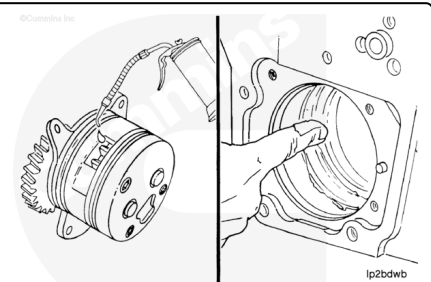
Install two new o-rings on the oil pump body.

Install the thickest o-ring into the groove nearest the oil pump drive gear.



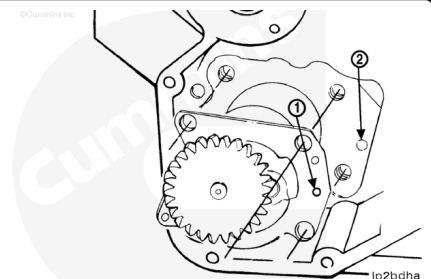
Use clean 15W-40 oil to lubricate the oil pump gears.

Apply a film of grease to the oil pump bore in the cylinder block.



⚠ CAUTION ⚠

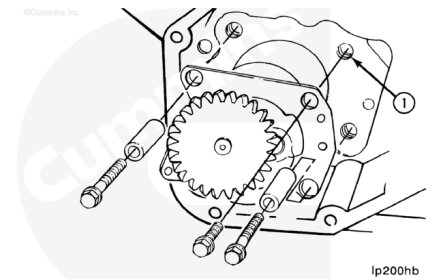
The dowel pin bore in the oil pump flange (1) must be aligned with the dowel pin in the cylinder block (2) to prevent damage to the oil pump flange during installation. Do not use the mounting capscrews to pull the oil pump into the bore. This can damage the mounting flange of the oil pump.



Install the oil pump by hand.

⚠ CAUTION ⚠

The shorter flange head capscrew must be used in the upper right-hand mounting hole (1) to prevent the idler gear from contacting the head of the capscrew and causing damage to the lubricating oil pump.



Sealant, Part Number 3824038, or equivalent, **must** be used to coat the threads of the three mounting capscrews to prevent air from being drawn past the capscrews during engine operation.

Install the three capscrews and spacers. Use the short flange head capscrew in the upper right hand mounting hole (1). Use the longer flange head capscrews and spacers in the other two holes.

Torque Value: 41 n•m [30 ft•lb]

Finishing Steps

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



- Install the water pump idler gear. Refer to Procedure 001-040 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-040-tr.html>)
- Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)
- Install the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)

- Operate the engine and check for leaks.

Last Modified: 23-Jan-2009
